openengine

Compare Otto and Diesel engines on P-V diagrams, in five sessions.

PAN IIT OSS explainer • higher secondary to diploma / undergraduate (levels 4-5)

Why this exists

Every engine turns heat into work and rejects the rest. This explainer follows the idea from the energy balance to the Otto, Diesel and Carnot cycles and a numerical simulation, using only the gas laws and the first law of thermodynamics.

The end goal

Compare Otto and Diesel cycle efficiency on P-V diagrams. The end goal is to compare Otto and Diesel cycle efficiency on P-V diagrams: to know what you hold equal, read the loops, and explain which engine wins and why. The same ideas apply to gas turbines, power plants and refrigerators.

How the five sessions build up

1. **Session 1: Heat in, work out, heat rejected** (Level 1, conceptual interactive model, 60 min).
2. **Session 2: Otto, Diesel and Carnot efficiency** (Level 2, mathematical engineering model, 75 min).
3. **Session 3: Simulate the cycle and validate it** (Level 3, simulation and validation, 90 min).
4. **Session 4: Temperature-dependent gas properties** (Level 2, mathematical engineering model, 75 min).
5. **Session 5: The dual cycle** (Level 2, mathematical engineering model, 75 min).

1 Session 1: Heat in, work out, heat rejected

A. Real-world problem

A car burns fuel, but most of the fuel's energy leaves through the exhaust and the radiator instead of turning the wheels. Why can no engine turn all the heat into work? And if you only see the pressure and volume of the gas inside the cylinder, how can you read off the work of one cycle? (General education only: this is a simplified model, not an engine design method.)

B. Concept and theory

An engine takes heat Qh from a hot source, does work W and must throw heat Qc away to a cold place. After one full cycle the engine is back where it started, so its stored energy has not changed and the energy balance is Qh = W + Qc. The efficiency is the part of the heat that becomes work: eta = W / Qh. The second law says Qc cannot be zero; between a hot source at Th and cold surroundings at Tc (kelvin), no engine can beat eta = 1 - Tc / Th. On a graph of pressure against volume, expansion at high pressure gives more work than compression at low pressure takes back, so a clockwise loop gives net work and the work is the enclosed area. With P in kPa and V in litres, kPa times L equals joules.

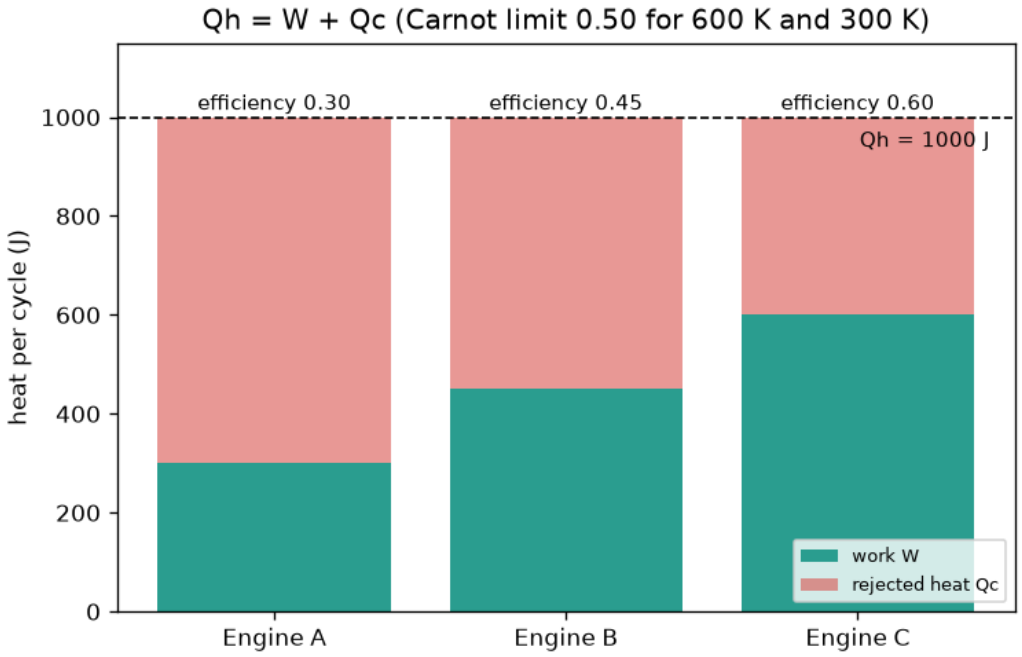
- Qh: heat taken from the hot source per cycle (J)
- Qc: heat rejected to the cold surroundings per cycle (J)
- W: net work per cycle (J)
- eta: thermal efficiency W / Qh (no unit)
- Th, Tc: hot and cold temperatures, absolute (K)
- P, V: gas pressure and volume (kPa, L)

$$Q_h = W + Q_c, \qquad W = Q_h - Q_c$$

$$\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

$$\eta \le \eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h}$$

$$W = \oint P\, dV$$



Three engines taking 1000 J: work and rejected heat; Engine C breaks the Carnot limit.

E. Worked example

An engine takes 1000 J from a 600 K source and rejects 650 J to 300 K surroundings. Find W and the efficiency and check the Carnot limit. Then find the work of a P-V rectangle from 100 to 300 kPa and 1 to 3 L.

1. Energy balance: W = Qh - Qc = 1000 - 650 = 350 J.
2. Efficiency = 350 / 1000 = 0.35.
3. Carnot limit = 1 - 300 / 600 = 0.50; since 0.35 is below 0.50 the engine is allowed.
4. Loop area = (300 - 100) kPa x (3 - 1) L = 200 x 2 = 400 kPa L = 400 J.
5. If that loop were the same engine's cycle with Qh = 1000 J, then Qc = 600 J and eta = 0.40.

Answer. W = 350 J, efficiency 0.35 (allowed, limit 0.50); the rectangle encloses 400 J.

F. Questions

1. **Conceptual.** Why can an engine never turn all of the heat it receives into work, even if it had no friction at all?
2. **Numerical.** An engine takes 800 J from a hot source and rejects 520 J. Find the work and the efficiency, and check the units and the answer is reasonable.
3. **Graph and simulation interpretation.** In the P-V view the loop is the same width but taller after you raise the work. By what factor does the area grow when the top pressure goes from 300 kPa to 500 kPa (bottom 100 kPa, volume 1 to 3 L)? What are the two works?
4. **Design and optimisation challenge.** Design: an engine must give 300 J of work per cycle between a 700 K source and 300 K surroundings. What is the smallest heat Qh it must take in, and the smallest Qc? A real engine reaches 0.30. How much heat would it need?

2 Session 2: Otto, Diesel and Carnot efficiency

A. Real-world problem

A petrol engine and a diesel engine both squeeze air, add heat and let the hot gas push a piston. What compression ratio do you need for an efficiency of 0.60, and why does a real engine, which is built to match such a cycle, reach much less? (Simplified air-standard model; not an engine design method.)

B. Concept and theory

The air-standard model treats the working fluid as an ideal gas (P v = R T per kg) with constant gamma = cp / cv = 1.4, heated from outside and without friction or heat loss. The Otto cycle has four steps: isentropic compression 1 to 2 by the compression ratio r = V1 / V2, heat addition at constant volume 2 to 3, isentropic expansion 3 to 4, and heat rejection at constant volume 4 to 1. Along an isentrope T V^(gamma-1) and P V^gamma stay constant, which gives T2 = T1 r^(gamma-1) and the efficiency 1 - 1 / r^(gamma-1). The Diesel cycle adds heat at constant pressure, so the volume grows by the cut-off ratio rho = V3 / V2 during burning; its loss 1 - eta is the Otto loss times a factor that is above 1 for rho above 1, so at the same r its efficiency is lower. A larger r raises both, with diminishing returns. The Carnot limit 1 - Tc / Th uses the highest and lowest temperatures: Otto and Diesel add heat over a range of temperatures, not all at the peak, so they stay well below it. Real engines are lower again because of finite burn time, heat loss, friction, gas leakage and the change of gas properties with temperature; typically quoted brake efficiencies of petrol and diesel engines are roughly 25 to 40 percent (check a data sheet for a specific engine).

- r: compression ratio V1 / V2 (no unit)
- rho: cut-off ratio V3 / V2 (Diesel) (no unit)
- gamma: cp / cv of the gas, 1.4 for air (no unit)
- T1..T4: gas temperature at states 1 to 4 (K)
- q_in, q_out: heat added, heat rejected (kJ/kg)
- R, cv: gas constant 0.287 and R / (gamma - 1) = 0.7175 (kJ/(kg K))

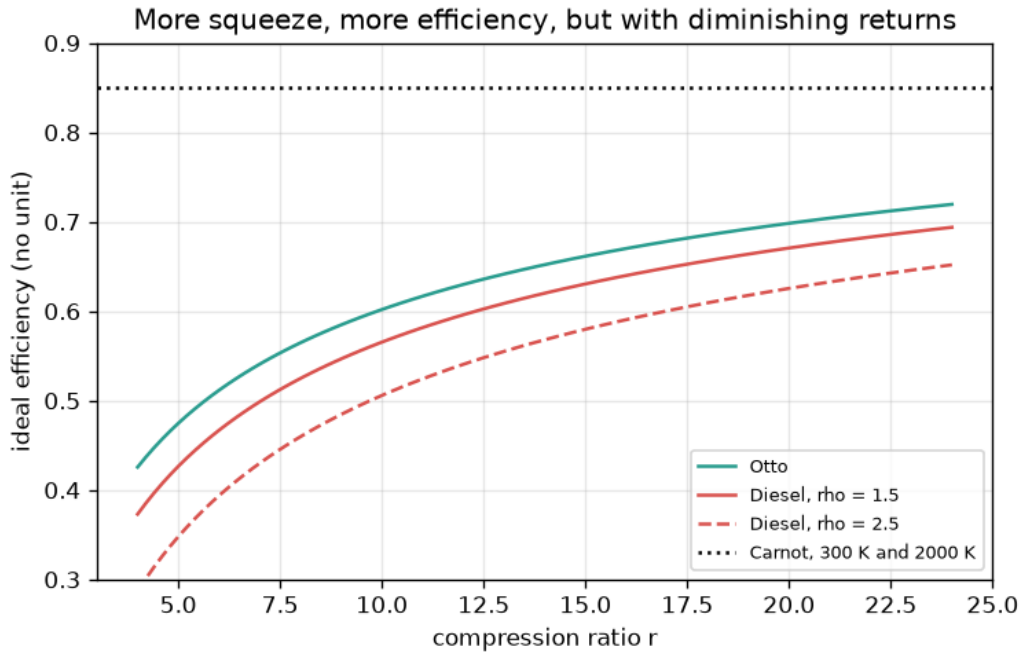
$$T_2 = T_1\, r^{\gamma-1}, \qquad PV^{\gamma} = \text{const (isentrope)}$$

$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}$$

$$\eta_{\text{Diesel}} = 1 - \frac{1}{r^{\gamma-1}} \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h}$$

$$r = (1 - \eta)^{-1/(\gamma-1)} \quad \text{(Otto design)}$$



Ideal efficiency against compression ratio for Otto, Diesel and the Carnot limit.

E. Worked example

An ideal Otto cycle with r = 9, gamma = 1.4, T1 = 300 K takes q_in = 1100 kJ/kg. Find the temperatures, the efficiency from the heats, and compare with the formula and the Carnot limit.

1. 9^0.4 = 2.408, so T2 = 300 x 2.408 = 722.5 K.
2. cv = 0.287 / 0.4 = 0.7175 kJ/(kg K). T3 = T2 + q_in / cv = 722.5 + 1100 / 0.7175 = 2255.6 K.
3. T4 = T3 / 2.408 = 936.6 K. Heat rejected q_out = cv (T4 - T1) = 0.7175 x 636.6 = 456.8 kJ/kg.
4. Efficiency from the heats = 1 - 456.8 / 1100 = 0.5848. Formula: 1 - 1 / 2.408 = 0.5847 (0.585). They agree.
5. Carnot between 300 K and 2255.6 K = 1 - 300 / 2255.6 = 0.867, much higher than 0.585, because heat is added from 722 K upward, not all at 2256 K.

Answer. T2 = 722.5 K, T3 = 2255.6 K, T4 = 936.6 K; efficiency 0.585 (Otto) against a Carnot limit of 0.867.

F. Questions

1. **Conceptual.** At the same compression ratio and the same heat input, which is more efficient, Otto or Diesel, and why?

2. **Numerical.** Find the ideal Otto efficiency for r = 8 and gamma = 1.4, and for the Diesel at r = 18 with rho = 2. Show the working.
3. **Graph and simulation interpretation.** From the efficiency graph, is it worth raising r from 19 to 20? Compare with raising it from 9 to 10.
4. **Design and optimisation challenge.** Design: choose r so that an ideal Otto engine reaches 0.60. A Diesel with r = 18 must also reach 0.60: what is the largest cut-off ratio allowed? Why will the real engines still be well below 0.60?

3 Session 3: Simulate the cycle and validate it

A. Real-world problem

The formulas of Session 2 assume that heat arrives instantly and that nothing is lost. A real engine burns fuel over tens of crank degrees, loses heat to its walls and drags against friction. Can we simulate the cycle step by step, show that it reproduces the formulas when we switch the real effects off, measure how much each effect costs, and choose a compression ratio for a small generator engine? (A simplified teaching model, not a design tool.)

B. Concept and theory

For a small step of a closed ideal gas the first law is dQ = dU + P dv with U = P v / (gamma - 1) per kg. This gives dP/dv = -gamma P / v with no heat (integrated with Runge-Kutta 4), dP = (gamma - 1) dQ / v at constant volume and dv = (gamma - 1) dQ / (gamma P) at constant pressure. The work is the sum of P dv around the closed loop (trapezoid rule), and the first law demands net work = q_in - q_out. A crank-angle model uses the slider-crank volume V(a) and releases heat along a Wiebe curve x = 1 - exp(-A z^(m+1)), z = (a - a0) / duration, with A = 6.908 and m = 2; the pressure obeys dP/da = ((gamma - 1) dQ/da - gamma P dV/da) / V. A share of the heat can be lost to the walls, and friction is subtracted as a friction mean effective pressure times the displacement. Two fair comparisons: at equal compression ratio and heat Otto wins, but at equal peak pressure (the structural limit) the Diesel can compress harder before burning and wins. Knock: in a spark-ignition engine the unburnt end gas, heated to about T1 r^(gamma-1), can ignite by itself, causing a damaging pressure shock; this sets an upper limit on r for a given fuel. A diesel compresses only air and injects fuel at the end, so it avoids this particular limit.

- v, P: specific volume and pressure of the gas (m3/kg, kPa)
- q_in: heat added per kg of air (kJ/kg)
- a: crank angle from top dead centre (degrees)
- loss: share of heat lost to the walls (no unit)
- fmep: friction mean effective pressure (kPa)
- T2: end-of-compression air temperature (K)

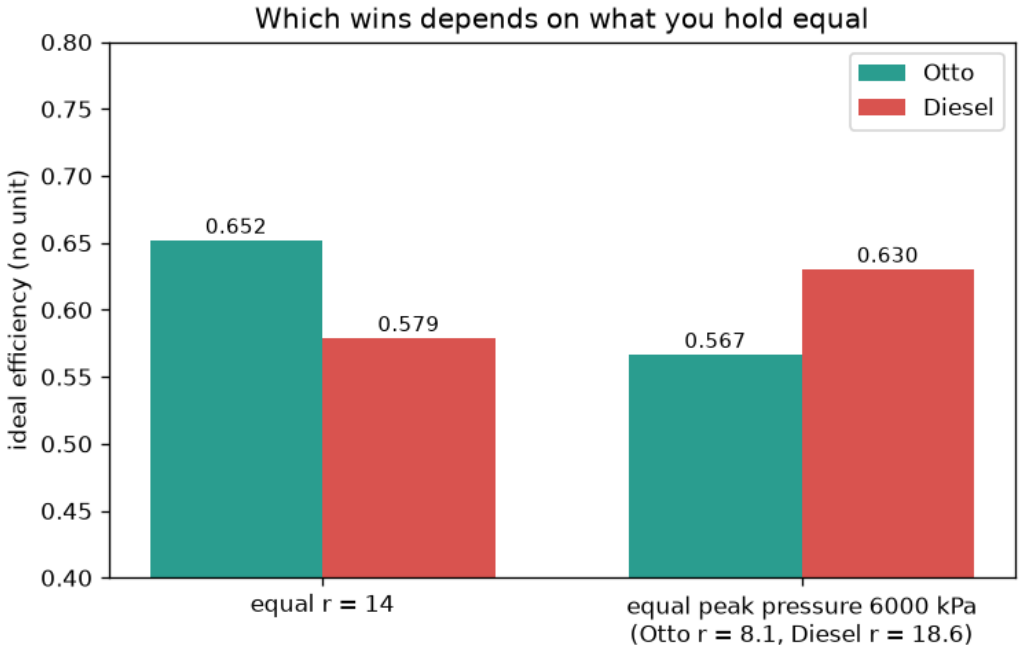
$$dQ = dU + P\,dv, \quad U = \frac{Pv}{\gamma - 1}$$

$$\frac{dP}{dv} = -\gamma\frac{P}{v} \text{ (no heat)}$$

$$\frac{dP}{da} = \frac{(\gamma - 1)\,dQ/da - \gamma P\,dV/da}{V}$$

$$x_b(a) = 1 - \exp\left[-A\left(\frac{a - a_0}{\Delta a}\right)^{m+1}\right]$$

$$W = \oint P\,dv \approx \sum \tfrac{1}{2}(P_i + P_{i+1})(v_{i+1} - v_i), \qquad W = q_{\text{in}} - q_{\text{out}}$$



Otto beats Diesel at equal compression ratio; Diesel beats Otto at equal peak pressure.

E. Worked example

Simulate the ideal Otto cycle with r = 9 and q_in = 1100 kJ/kg (P1 = 100 kPa, T1 = 300 K). Compare its efficiency with the formula and check the first law.

1. Compression: integrate dP/dv = -gamma P / v from v1 = 0.861 to v2 = 0.0957 m3/kg in 400 steps.
2. Heat at constant volume: add 1100 kJ/kg in 400 equal steps, dP = 0.4 dQ / v2; then expand back to v1 and cool to P1 at constant volume.
3. Sum P dv around the loop: net work = 643.3 kJ/kg; heat rejected = v1 (P4 - P1) / 0.4 = 456.8 kJ/kg.
4. Efficiency = 643.3 / 1100 = 0.58477; the formula gives 0.58476.
5. First law: net work - (q_in - q_out) = 0.02 kJ/kg, about 0.002 percent of the heat input.

Answer. Simulated efficiency 0.5848 (formula 0.5848), net work 643 kJ/kg, rejected heat 457 kJ/kg, first-law error 0.02 kJ/kg.

F. Questions

1. **Conceptual.** Why does a Diesel win at equal peak pressure although an Otto wins at equal compression ratio?
2. **Numerical.** Find the end-of-compression temperature for r = 10 and r = 9 (T1 = 300 K, gamma = 1.4). A fuel allows T2 up to 740 K with a 20 K margin: is r = 10 acceptable?

3. **Graph and simulation interpretation.** In the knock-down figure, which single real effect costs the most efficiency, and by how much? Compare the burn time, the heat loss and the friction.
4. **Design and optimisation challenge.** Design: a generator engine uses fuel A (T2 limit 740 K) with a 20 K margin, or fuel B (800 K). Choose r for each in steps of 0.5, find the modelled brake efficiency (10 percent loss, friction 100 kPa) and say which fuel meets a 0.45 target. What r would be needed?

4 Session 4: Temperature-dependent gas properties

A. Real-world problem

Session 2 promised an ideal Otto efficiency of 0.585 at r = 9, yet a real petrol engine reaches perhaps 0.25 to 0.35. Session 3 blamed burn time, heat loss and friction. A quieter part of the gap hides in one number: gamma = 1.4 is the value for cold air. When air is hot its molecules vibrate as well as move and spin, so each kelvin costs more energy and gamma falls. How much efficiency does that cost even in a perfect engine, and does it change the best compression ratio? (Simplified air-standard model; not an engine design method.)

B. Concept and theory

An ideal gas still obeys P v = R T, but its heat capacities depend on temperature. Data tables (JANAF, NASA) fit cp of each gas with a polynomial cp / R = a1 + a2 T + a3 T^2 + a4 T^3 + a5 T^4, with one set of coefficients for 200 to 1000 K and another for 1000 to 6000 K. Air is 79 percent N2 and 21 percent O2 by moles, and for a mixture of ideal gases the coefficients mix in that ratio. Then cv = cp - R and gamma = cp / cv, which falls from 1.40 at 300 K to 1.33 at 1000 K and 1.30 at 2000 K. Integrating cp gives the enthalpy h(T) and the internal energy u = h - R T. Heat at constant volume raises u, so q_in = u(T3) - u(T2) and q_out = u(T4) - u(T1). For a reversible step with no heat the first law gives cv(T) dT / T = -R dv / v, which no longer integrates to a power law. Instead define phi(T) = s0(T) - R ln T, where s0 is the tabulated standard entropy: then phi(T2) - phi(T1) = R ln(v1 / v2) = R ln r. Because phi always grows with T, each state follows from a one-line bisection. With constant gamma, u = cv T and phi = cv ln T, and every step reduces to Session 2. With variable cp the compression ends a little cooler, the same heat raises the temperature much less because hot air has a larger cv, and the expansion with its low gamma returns less of the heat as work. A useful summary is the effective gamma: the constant value that would give the same efficiency in the Otto formula.

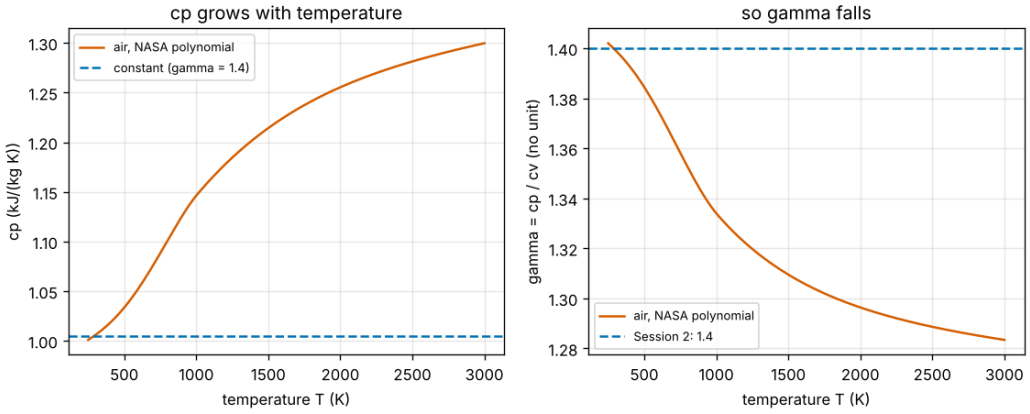
- cp, cv: specific heats at constant pressure and volume (kJ/(kg K))
- gamma: cp / cv at a given temperature (no unit)
- a1..a7: NASA polynomial coefficients (a6, a7 fix h and s) (various)
- u, h: specific internal energy and enthalpy (kJ/kg)
- s0, phi: standard entropy at 1 bar, and phi = s0 - R ln T (kJ/(kg K))
- gamma_eff: constant gamma giving the same Otto efficiency (no unit)

$$\frac{c_p(T)}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4, \qquad c_v = c_p - R, \qquad \gamma = \frac{c_p}{c_v}$$

$$u(T) = h(T) - RT, \qquad q_{\text{in}} = u(T_3) - u(T_2), \qquad q_{\text{out}} = u(T_4) - u(T_1)$$

$$c_v(T)\,\frac{dT}{T} = -R\,\frac{dv}{v} \quad \Rightarrow \quad \phi(T_2) - \phi(T_1) = R\ln r, \qquad \phi(T) = s^0(T) - R\ln T$$

$$\eta = 1 - \frac{q_{\text{out}}}{q_{\text{in}}}, \qquad \gamma_{\text{eff}} = 1 + \frac{\ln(1/(1-\eta))}{\ln r}$$



cp of air rises with temperature and gamma falls; dashed lines are the constant Session 2 values.

E. Worked example

An ideal Otto cycle with r = 9, T1 = 300 K and q_in = 1100 kJ/kg (the Session 2 example), but with cp(T) of air from the NASA polynomial. Find T2, T3, T4, the efficiency and the effective gamma, and check the first law.

1. Compression: phi(300 K) = 5.0787 kJ/(kg K) and R ln 9 = 0.287 x 2.1972 = 0.6306, so phi(T2) = 5.7093. Bisection gives T2 = 700.0 K (constant gamma gave 722.5 K).
2. Heat at constant volume: u(T2) = 215.4 kJ/kg, so u(T3) = 215.4 + 1100 = 1315.4 kJ/kg and T3 = 1927 K (constant gamma gave 2256 K). The mean cv over this step is 1100 / 1227 = 0.896 kJ/(kg K), not 0.7175.
3. Expansion: phi(T4) = phi(T3) - 0.6306, so T4 = 964 K (constant gamma gave 937 K).
4. Rejected heat: q_out = u(964 K) - u(300 K) = 432.4 - (-84.2) = 516.6 kJ/kg (constant gamma gave 456.8). Efficiency = 1 - 516.6 / 1100 = 0.530.
5. Effective gamma = 1 + ln(1 / 0.470) / ln 9 = 1.344, between gamma(T3) = 1.298 and gamma(T1) = 1.399. The P dv area of the loop traced with RK4 is 583.4 kJ/kg, equal to q_in - q_out = 583.4 kJ/kg.

Answer. T2 = 700 K, T3 = 1927 K, T4 = 964 K; efficiency 0.530 against 0.585 with constant gamma, below the roadmap target of 0.56. The effective gamma is 1.344.

F. Questions

1. **Conceptual.** Why does adding more heat per cycle lower the ideal Otto efficiency once cp depends on temperature, while in Session 2 the efficiency did not depend on the heat at all?
2. **Numerical.** Using `cp_air`, find cp, cv and gamma of air at 1500 K. Then estimate the Otto efficiency at r = 9 if you used that single gamma for the whole cycle. Is it above or below the variable-cp answer of 0.530, and why?
3. **Graph and simulation interpretation.** In the left panel of the efficiency figure, the variable-cp curve approaches a value near the constant-gamma line as the heat input goes to zero, but not exactly 0.585. Explain why it starts there and why it does not quite reach 0.585.

4. **Design and optimisation challenge.** Design: with Session 2's constant gamma, r = 9.88 gives an ideal efficiency of 0.60. With variable cp and q_in = 1100 kJ/kg, what compression ratio is needed for 0.60, and what is T2? If the fuel knocks above an end-of-compression temperature of about 750 K (an illustrative number), is this design possible? What would you trade?

5 Session 5: The dual cycle

A. Real-world problem

Session 2 had two extremes: the Otto cycle adds all its heat at constant volume, the Diesel cycle all of it at constant pressure. A real high-speed diesel does both. The fuel injected just before top dead centre mixes with hot air and burns almost at once, at nearly constant volume; the rest burns as it is injected while the piston moves down, at nearly constant pressure. How does the share of heat added at constant volume change the efficiency, and why do engine designers not simply add all of it at constant volume? (Simplified air-standard model; not an engine design method.)

B. Concept and theory

The dual cycle, also called the Seiliger or Sabathe cycle, has five states. 1 to 2 is isentropic compression by the compression ratio r. 2 to 3 adds a share x of the heat at constant volume, raising the pressure by the pressure ratio alpha = P3 / P2 = T3 / T2. 3 to 4 adds the rest, (1 - x) q_in, at constant pressure, raising the volume by the cut-off ratio rho = v4 / v3 = T4 / T3. 4 to 5 is isentropic expansion back to the largest volume, and 5 to 1 rejects heat at constant volume. Using cv for the first heat and cp = gamma cv for the second, and q_out = cv (T5 - T1), the efficiency simplifies to the closed form below. Two checks show that it contains Sessions 2's formulas: with rho = 1 (x = 1) the last fraction is (alpha - 1) / (alpha - 1) = 1 and the Otto formula returns; with alpha = 1 (x = 0) it becomes (rho^gamma - 1) / (gamma (rho - 1)), the Diesel factor. At the same r and heat, the heat added at constant volume is expanded through the whole stroke while the constant-pressure heat is expanded less, so the efficiency rises steadily from the Diesel value to the Otto value as x grows. The price is the peak pressure P3 = alpha P2, which also rises with x and must be carried by the cylinder head, the piston and the bearings.

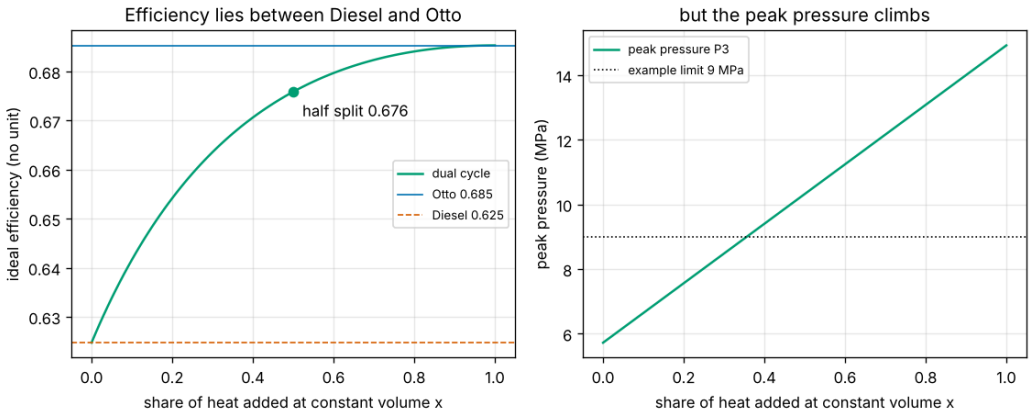
- r: compression ratio v1 / v2 (no unit)
- x: share of the heat added at constant volume (no unit)
- alpha: pressure ratio P3 / P2 during constant-volume heating (no unit)
- rho: cut-off ratio v4 / v3 during constant-pressure heating (no unit)
- T1..T5, P1..P5: temperature and pressure at the five states (K, kPa)
- q_in, q_out: heat added, heat rejected (kJ/kg)

$$x\,q_{\text{in}} = c_v(T_3 - T_2), \qquad (1-x)\,q_{\text{in}} = c_p(T_4 - T_3)$$

$$\alpha = \frac{P_3}{P_2} = \frac{T_3}{T_2}, \qquad \rho = \frac{v_4}{v_3} = \frac{T_4}{T_3}$$

$$\eta_{\text{dual}} = 1 - \frac{1}{r^{\gamma-1}}\,\frac{\alpha\rho^{\gamma} - 1}{(\alpha - 1) + \gamma\alpha(\rho - 1)}$$

$$\rho = 1:\ \eta = \eta_{\text{Otto}}, \qquad \alpha = 1:\ \eta = \eta_{\text{Diesel}}$$



Efficiency of the dual cycle against the constant-volume share, between Diesel and Otto (left), and its peak pressure (right).

E. Worked example

A dual cycle with r = 18, T1 = 300 K, P1 = 100 kPa takes q_in = 1100 kJ/kg, half at constant volume and half at constant pressure. Find the temperatures, alpha, rho, the peak pressure and the efficiency, and check the result with the closed-form formula.

1. Compression: 18^0.4 = 3.1777, so T2 = 300 x 3.1777 = 953.3 K and P2 = 100 x 18 x 3.1777 = 5719.8 kPa.
2. Constant volume: T3 = 953.3 + 550 / 0.7175 = 953.3 + 766.6 = 1719.9 K, so alpha = 1719.9 / 953.3 = 1.804 and the peak pressure P3 = 1.804 x 5719.8 = 10319 kPa.
3. Constant pressure: T4 = 1719.9 + 550 / 1.0045 = 1719.9 + 547.5 = 2267.4 K, so rho = 2267.4 / 1719.9 = 1.318.
4. Expansion from rho v2 to r v2: T5 = 2267.4 x (1.318 / 18)^0.4 = 2267.4 x 0.3515 = 796.9 K. Heat rejected q_out = 0.7175 x (796.9 - 300) = 356.6 kJ/kg, so eta = 1 - 356.6 / 1100 = 0.676.
5. Formula: alpha rho^1.4 - 1 = 1.804 x 1.4725 - 1 = 1.6565 and (alpha - 1) + 1.4 alpha (rho - 1) = 1.6082, ratio 1.0300; eta = 1 - 0.3147 x 1.0300 = 0.676. It agrees and lies between the Diesel value 0.625 and the Otto value 0.685.

Answer. T2 = 953 K, T3 = 1720 K, T4 = 2267 K, T5 = 797 K; alpha = 1.804, rho = 1.318, peak pressure 10.3 MPa; efficiency 0.676, between Diesel (0.625) and Otto (0.685).

F. Questions

1. **Conceptual.** At the same r and the same heat, why does moving heat from constant pressure to constant volume raise the efficiency, and why is that not free?
2. **Numerical.** A dual cycle has r = 16, q_in = 1000 kJ/kg and x = 0.4 (T1 = 300 K, gamma = 1.4). Find T2, T3, T4, alpha, rho and the efficiency.
3. **Graph and simulation interpretation.** In the efficiency figure the curve rises quickly for small x and flattens near x = 1, while the peak pressure rises in a straight line. What does that mean for the last 10 percent of heat moved to constant volume?
4. **Design and optimisation challenge.** Design: the cylinder head of a diesel engine allows a peak pressure of 9000 kPa. With q_in = 1100 kJ/kg, find the largest constant-volume share x and the resulting efficiency at r = 16, 18 and 20. Which r would you choose, and what limits r from above?

6 Real-world mini-project

Choose the compression ratio of a small generator engine. A small generator engine can run on fuel A or fuel B. For illustration, fuel A tolerates an end-of-compression air temperature T2 of 740 K before it knocks and fuel B tolerates 800 K; the designer keeps a margin of 20 K. Use `max_r_knock_free` and `engine_cycle` from Session 3 (heat 1100 kJ/kg of air, Wiebe burn of 40 degrees from -10, 10 percent heat loss, friction mean effective pressure 100 kPa). The owner wants a brake efficiency of at least 0.45. These limits are illustrative numbers, not fuel specifications.

7 The path ahead

1. **Session 6: T-S diagram from the simulation.** Compute entropy along the simulated cycle and draw the T-S loop with the Carnot rectangle.
2. **Session 7: A knock model with engine speed.** An induction-time integral on the end-gas temperature history.
3. **Session 8: Four strokes with a pumping loop.** Add gas exchange to the crank-angle model with a throttled intake.

8 Answers and hints

Session 1

1. **Conceptual.** Hint: What does the second law say about Qc, and what would eta = 1 - Tc / Th need? Solution: The second law forbids a cyclic engine that takes heat from one source and turns all of it into work. Some heat Qc must go to a colder place. The best case is the Carnot limit 1 - Tc / Th, which reaches 1 only if Tc were 0 K (or Th infinite), and neither is possible. So even a frictionless engine has an efficiency below 1.
2. **Numerical.** Hint: W = Qh - Qc, then divide by Qh. Solution: W = 800 - 520 = 280 J. eta = 280 / 800 = 0.35 (no unit). Sanity check: the efficiency is between 0 and 1, and W plus Qc gives back 800 J.
3. **Graph and simulation interpretation.** Hint: Area = (top - bottom) x width. Solution: The heights are 200 kPa and 400 kPa, so the area doubles: 400 J becomes 800 J (units: kPa L = J).
4. **Design and optimisation challenge.** Hint: Use the Carnot limit as the best case: Qh = W / eta. Solution: Best case eta = 1 - 300 / 700 = 0.571, so Qh = 300 / 0.571 = 525 J and Qc = 225 J. A real engine at 0.30 needs Qh = 300 / 0.30 = 1000 J and rejects 700 J. Trade-off: the Carnot value is a floor on the heat you must supply, not a design target, because real engines are far below it.

Session 2

1. **Conceptual.** Hint: Where in the piston stroke is the heat added in each cycle? Solution: Otto. Its heat is added at constant volume while the piston is at the top, so all of it is expanded through the full stroke. The Diesel adds part of its heat while the piston is already moving down, so that heat is expanded less. The factor (rho^gamma - 1) / (gamma (rho - 1)) is above 1, which lowers the Diesel efficiency.
2. **Numerical.** Hint: Use 8^0.4 and 18^0.4, and 2^1.4 = 2.639. Solution: Otto: 8^0.4 = 2.297, so eta = 1 - 1 / 2.297 = 0.565. Diesel: 18^0.4 = 3.178, so 1 / 3.178 = 0.3147; factor = (2.639 - 1) / (1.4 x 1) = 1.171; eta = 1 - 0.3148 x 1.171 = 0.632. Both are between 0 and 1 and below the Otto value at r = 18 (0.685), as expected.

3. **Graph and simulation interpretation.** Hint: Read how steep each curve is at those r values. Solution: The curves flatten. From 9 to 10 the Otto efficiency gains 0.017; from 19 to 20 only 0.006, about a third as much, while the cost (peak pressure, knock, strength) rises. So the same extra squeeze buys much less at high r.
4. **Design and optimisation challenge.** Hint: Invert: r = (1 - eta)^(-1/0.4); for the Diesel solve the formula for rho numerically. Solution: Otto: r = 0.4^(-2.5) = 9.88, about 10. Diesel at r = 18: the largest rho is 2.73 (above that eta drops below 0.60); in heat terms this limits how much fuel can burn at constant pressure. Real engines are lower because the burn takes time, heat leaks to the walls, friction takes work, and the gas properties change at high temperature (gamma falls); the model also ignores gas exchange. Constraint: higher r raises pressure, structural load and (for petrol) the risk of knock, treated in Session 3.

Session 3

1. **Conceptual.** Hint: What limits each engine: the compression ratio or the peak pressure? Solution: At equal r the comparison favours the Otto because all its heat is added at top dead centre. But strength limits the peak pressure, not r. The Otto's peak pressure comes after heat addition at constant volume, so it must keep r low (8.1 for 6000 kPa); the Diesel's peak pressure is the end of compression, so it can use r = 18.6 and the higher r wins (0.63 against 0.57).
2. **Numerical.** Hint: T2 = T1 r^0.4; margin means T2 must not exceed 720 K. Solution: r = 10: 10^0.4 = 2.512, so T2 = 300 x 2.512 = 754 K. r = 9: 9^0.4 = 2.408, so T2 = 722 K. The allowed value is 740 - 20 = 720 K, so r = 10 is not acceptable and even r = 9 (722 K) is just too hot; the largest allowed r in steps of 0.5 is 8.5 (T2 = 706 K).
3. **Graph and simulation interpretation.** Hint: Take differences between the neighbouring bars: 0.584, 0.570, 0.513, 0.443. Solution: Burn time costs 0.584 - 0.570 = 0.014, 10 percent heat loss costs 0.570 - 0.513 = 0.057 and friction of 100 kPa costs 0.513 - 0.443 = 0.070. Friction costs the most in this setting, then heat loss; the finite burn is the smallest. (The numbers depend on the illustrative values.)
4. **Design and optimisation challenge.** Hint: Use max_r_knock_free and engine_cycle. Solution: Fuel A: r = 8.5 (T2 = 706 K), brake 0.436: misses the target. Fuel B: r = 10.5 (T2 = 768 K), brake 0.463: meets it. The smallest r that reaches 0.45 is 9.5 (T2 = 738 K), which fuel A cannot allow with its margin. Trade-off: a better fuel (or a lower friction or loss) is needed for the target; raising r above the knock limit would risk damage. All limits are illustrative.

Session 4

1. **Conceptual.** Hint: What does more heat do to T3, and what does a higher temperature do to gamma during the expansion? Solution: With constant gamma the efficiency 1 - 1 / r^(gamma-1) contains only r and gamma, so the heat cancels out. With variable cp, more heat means a hotter gas, and hot air has a larger cv and a smaller gamma. The expansion follows an isentrope with low gamma, along which the temperature falls less for the same volume ratio, so the gas leaves the expansion hotter and more heat is thrown away at the bottom. The more heat, the hotter the cycle and the lower the effective gamma: 1.364 at 500 kJ/kg, 1.344 at 1100 kJ/kg, 1.326 at 2000 kJ/kg.
2. **Numerical.** Hint: cv = cp - 0.287; eta = 1 - 9^(1 - gamma). Solution: cp(1500 K) = 1.214 kJ/(kg K), cv = 1.214 - 0.287 = 0.927, gamma = 1.214 / 0.927 = 1.310. Then eta = 1 - 9^(-0.310) = 1 - 0.506 = 0.494. This is below 0.530 because the real cycle also spends time at low temperature (compression from 300 K to 700 K) where gamma is close to 1.4. A single hot gamma is too pessimistic and a single cold gamma (0.585) too optimistic; the effective value 1.344 lies in between.

3. **Graph and simulation interpretation.** Hint: With very little heat, the cycle only spans the compression from 300 K to T2. What is gamma there? Solution: With almost no heat added the expansion retraces the compression, so the cycle only sees temperatures from 300 K to about 700 K. There gamma falls from 1.40 to about 1.36, so the effective gamma is a little below 1.40 and the efficiency (about 0.57) is a little below 0.585. As the heat grows, the cycle reaches temperatures where gamma is lower still and the curve falls further.
4. **Design and optimisation challenge.** Hint: Bisection on r with `otto_cycle(r, 1100)["eta"]`, then read `["T"][1]`. Solution: Bisection gives r = 14.4, with T2 = 829 K. That is well above the illustrative 750 K knock limit, so a spark-ignition engine on that fuel cannot reach 0.60 even in the ideal model. Trade-offs: accept a lower efficiency at a knock-safe r (about 0.53 at r = 9); use a fuel with a higher knock limit; or move to compression ignition (a diesel compresses only air, so r = 14 to 20 is allowed), accepting higher peak pressures and the constant-pressure heat addition of Session 2. The lesson: the constant-gamma formula asks for too small an r and promises too much.

Session 5

1. **Conceptual.** Hint: How much of the stroke does each part of the heat get to expand through? What happens to P3? Solution: Heat added at constant volume is added at the smallest volume, so the hot gas expands through the full compression ratio r and turns more of that heat into work. Heat added at constant pressure is added while the volume is already growing by rho, so it is expanded through only r / rho and gives less work. More constant-volume heat therefore raises the efficiency towards the Otto value. The cost is the peak pressure P3 = alpha P2, which rises with every extra kJ added at constant volume (from 5.7 MPa at x = 0 to 14.9 MPa at x = 1 for r = 18), and the engine structure must carry it.
2. **Numerical.** Hint: T2 = 300 x 16^0.4; then 400 kJ/kg with cv = 0.7175 and 600 kJ/kg with cp = 1.0045. Solution: 16^0.4 = 3.031, so T2 = 909.4 K. T3 = 909.4 + 400 / 0.7175 = 1466.9 K, alpha = 1.613. T4 = 1466.9 + 600 / 1.0045 = 2064.2 K, rho = 1.407. T5 = 2064.2 x (1.407 / 16)^0.4 = 780.6 K, q_out = 0.7175 x 480.6 = 344.9 kJ/kg, eta = 1 - 344.9 / 1000 = 0.655. Check: between Diesel and Otto at r = 16 (Otto 1 - 1 / 3.031 = 0.670).
3. **Graph and simulation interpretation.** Hint: Compare the efficiency gained from x = 0 to 0.1 with the gain from 0.9 to 1.0, and the pressure rise in each step. Solution: Each step of x raises the peak pressure by the same amount (about 0.92 MPa per 0.1 at r = 18 and 1100 kJ/kg), but the efficiency gain shrinks: the first tenth gains 0.017, the last tenth only 0.0003. So the last bit of constant-volume heat costs the most pressure for the least efficiency. That is why real diesels aim for a moderate pressure rise rather than a pure Otto-like burn.
4. **Design and optimisation challenge.** Hint: x_max = cv T2 (P_max / P2 - 1) / q_in; use `max_cv_share` and `dual_cycle`. Solution: r = 16: x = 0.508, eta = 0.660. r = 18: x = 0.357, eta = 0.668. r = 20: x = 0.232, eta = 0.672. A higher r leaves less room for constant-volume heat because P2 is already higher, but it still wins slightly, with diminishing returns. Above about r = 24.9 the compression alone (P2 = 100 r^1.4) exceeds 9000 kPa and no heat at all can be added at constant volume. Real choices also weigh friction, which grows with r and pressure, heat loss and cold starting; the model gives the direction, not the final number.

9 How to join

Read `CONTRIBUTING.md` in the repository, then pick an issue labelled `good first issue`. Repository: https://github.com/KrunalPanchal1995/pan-iit-oss/tree/main/openengine